Transformer-Enhanced Multi-Agent Reinforcement Learning for Adaptive Honeypot Deception

1. Introduction and Problem Statement

Modern cyber attackers increasingly use automated tools to fingerprint and evade static defensive measures. Deception technologies, like honeypots, are crucial for proactive defense, but their current static nature—presenting the same predetermined vulnerabilities and responses—makes them detectable by sophisticated adversaries. While deploying honeypots in swarms improves coverage, it exacerbates the management problem without addressing the fundamental lack of adaptability.

This thesis proposes a novel solution by creating an *adaptive honeypot swarm* that uses Artificial Intelligence to dynamically tailor its deceptive responses to an attacker's behavior in real-time. The core innovation is the integration of a Transformer model for understanding attacker intent with Multi-Agent Reinforcement Learning (MARL) for coordinating deceptive actions across the swarm. This enables the system to learn complex engagement strategies, prolonging attacker interaction and maximizing intelligence gathering.

2. Research Objectives and Methodology

The primary goal is to develop and evaluate an AI-driven honeypot framework that outperforms static deception systems by intelligently modeling attacker behavior and adapting accordingly.

1. Develop a Simulation Environment for Honeypot-Attacker Interaction.

- **Methodology:** Create a configurable network simulation in Python. The environment will feature a swarm of 3-5 honeypot agents and a scripted "red team" attacker based on the MITRE ATT&CK framework. The attacker will perform multi-stage attacks (reconnaissance, exploitation, lateral movement).
- **Outcome:** A reproducible testbed for training and evaluating AI-based deception policies.

2. Design and Pre-train a Transformer-based Attacker Behavior Encoder.

- **Methodology:** Implement a Transformer encoder model. It will be pre-trained on synthetic sequences of attacker actions in a self-supervised manner (e.g., masked token prediction) to learn the underlying patterns of offensive workflows.
- **Integration:** During MARL training, this Transformer will process the recent history of attacker actions and produce a contextualized embedding vector. This embedding becomes a critical part of the state representation for each RL agent, providing temporal context and intent inference.
- **Novelty:** This moves beyond simple state representations, allowing the system to reason about the *sequence and context* of an attack.

3. Implement a Centralized-Training-Decentralized-Execution (CTDE) MARL Framework.

- **Methodology:** Use a proven MARL algorithm (e.g., MADDPG or QMIX). The central critic will be trained using experiences from all honeypots, incorporating the Transformer's context embeddings. Each honeypot will then execute its own policy based on its local observations and the shared context.
- **Action Space:** Honeypot actions will include dynamic deceptions: changing service banners, altering filesystem structures, and controlling the pace of fake data "leaks."
- **Reward Function:** The reward will incentivize sustained attacker engagement (positive reward per step), deeper penetration into deceptive scenarios (bonus rewards), and penalize attacker disconnection.

4. Conduct a Comparative Evaluation Against Strong Baselines.

- **Methodology:** Run extensive experiments in the simulation environment, comparing the proposed Transformer-MARL framework against:
    - **Baseline 1:** Static honeypots (conventional approach).
    - **Baseline 2:** MARL honeypots without the Transformer (using a simple LSTM or sliding-window state).
- **Primary Metrics:**
    - **Attacker Dwell Time:** The total number of interaction steps.

- **Interaction Depth:** The number of unique MITRE ATT&CK techniques performed.
- **Generalization:** Performance against a novel, unseen attack script after training.

3. Expected Contribution and Novelty

This thesis will deliver a novel, practical integration of Transformers and MARL for cybersecurity deception. The key contributions are:

1. **A Novel Architecture:** The proposed Transformer-MARL framework is a new approach to adaptive cyber defense, demonstrating how sequence modeling can enhance decision-making in dynamic environments.
2. **A Public Benchmark:** The simulation environment and baseline results will be released publicly, providing a valuable benchmark for future research in AI-driven security.
3. **Empirical Evidence:** The work will provide rigorous evidence that modeling attacker behavior sequences leads to more effective and robust deception policies compared to static or non-sequential AI approaches.

4. Technical Skills and Feasibility

Skill set needed:

- **Core ML/DL:** Python, PyTorch/TensorFlow, neural network training.
- **Reinforcement Learning:** Understanding of RL concepts and libraries (e.g., Stable-Baselines3).
- **Software Engineering:** Git, object-oriented programming, and building simulations. The student will deepen their expertise in cutting-edge Transformers and Multi-Agent Systems, making this thesis an excellent portfolio piece.

5. Project Timeline

- **Months 1-2:** Literature Review & Development of Simulation Environment.
- **Months 3-4:** Implementation and Pre-training of the Transformer Encoder.
- **Months 5-6:** Integration of Transformer with MARL Framework and Initial Training.
- **Months 7-8:** Comprehensive Evaluation, Ablation Studies, and Thesis Writing.